ICS & AI Transparency Log

Project 2 (PDE BVP) — Team Q

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Appended to the end of the Project 2 report. This document contains the team AI transparency log (Part A) and the individual contribution statement of Meichen Ding (Part B).

Part A is the team AI transparency log (one copy per team, filled in collectively); **Part B** is the individual contribution statement of Meichen Ding (one copy per member).

1 AI Transparency Log

This appendix records the distance between raw generative-AI output and the claims that survived into this report; it is a record of auditing, not a count of prompts. Every claim listed in A2 was re-derived, re-run against the team's own code, or falsified before it was allowed into the text, and each of the three audits changed a claim that an earlier draft had already stated. The conversation transcripts are retained by the team and can be supplied on request; they are not appended here.

A1. AI tools used

Table 1: Generative-AI tools used in Project 2. Entries describe the use that actually reached the report; routine autocomplete and spelling checks are covered by the last row.

Tool	Version / date	Primary use
DeepSeek-based assistant in DeepSeek Harness (`deepseek-flash`)	current build, Oct–Nov 2026	Drafting and re-deriving §1.2 and §2.1–§2.4 (model, weak form, truncation error, interface conductance, boundary elimination, discrete structure); writing the independent verification scripts in `support/verify/`; adversarial review of the section text.
Claude (Anthropic), web application	current build, Oct–Nov 2026	Drafting of the algorithm section, integration notes for merging the section files, figure and table captions, wording and consistency passes.
ChatGPT (OpenAI) and Gemini (Google)	current builds, Oct–Nov 2026	LaTeX error diagnosis, document structure, phrasing and completeness checks.
All of the above, routine use	—	Autocomplete, spelling, reference formatting, table alignment. No numerical value was accepted without reproduction (see A3).

A2. Significant AI-assisted claim audits

Three interactions are recorded below because each one changed a mathematical or numerical claim rather than simply rewording it. The common failure pattern is a claim that is true for the face coefficient, or for one dimension, being silently promoted to the solution of the two-dimensional problem.

Table 2: Claim audits. Column 3 states what would have shown the claim to be wrong.

Item	Prompt / claim / assumptions	Risk and possible falsifier
1	"Arithmetic averaging of the interface conductance produces an $\mathcal{O}(1)$ error in the interface flux, so the arithmetic-averaged solution is $\mathcal{O}(1)$ inaccurate." **Assumption hidden in the wording:** the face-coefficient error and the error of the discrete solution are the same object.	The claim mixes three different quantities: the relative face-conductance error $(1-c)^2/(4c)$ (genuinely $\mathcal{O}(1)$ in h), the nodal error and the midline interface-flux error (first order in h), and the *maximum* interface-flux error over the square, which is $\mathcal{O}(1)$ for a different reason — the exact Dirichlet trace on $y = 0, 1$. **Falsifier:** a refinement study. If the nodal error did not halve under mesh refinement, the first-order part of the claim would be wrong.

Table 2: Claim audits (continued).

Item	Prompt / claim / assumptions	Risk and possible falsifier
2	"Since $A_h \succeq a_{\min} A_h^{(1)}$ in the Loewner order and both matrices are M-matrices, the inverses compare entrywise, $A_h^{-1} \le a_{\min}^{-1}(A_h^{(1)})^{-1}$; hence $\|A_h^{-1}\|_\infty \le 1/(8a_{\min})$." **Assumption hidden:** spectral order implies entrywise order.	The implication is false: monotonicity of M-matrix inverses needs an *entrywise* hypothesis, while growing a makes the off-diagonals $-a_f/h^2$ *more negative*. **Falsifier:** evaluate both inverses for a strongly varying coefficient, e.g. $a = 1 + 9\sin^2(\pi x)$ at $n = 63$, where the entrywise ratio reaches $3.10 > 1$. The conclusion $\|A_h^{-1}\|_\infty \le 1/(8a_{\min})$ is nevertheless true and is therefore proved by a maximum-principle barrier argument instead.
3	"The arithmetic operator is symmetric and its shared face flux is conservatively balanced, so the interface physics is correct."	Symmetry and local conservation are necessary but not sufficient; they are satisfied by a matrix whose interface flux is wrong by 36%. **Falsifier:** compare the computed interface flux with the exact $J_* = -20/11$ on a benchmark whose exact solution is known. It differs by 0.662 while the two-sided reconstructed jump is 3.85×10^{-13} and the symmetry defect is zero.

A3. Independent verification and decision

Table 3: What was checked, where the evidence lives, and what changed.

Item	Independent check and evidence location	Final decision / modification / limitation
1	Re-derived the truncated-series argument in one dimension, where the discrete system reduces to a tridiagonal chain with a closed-form flux; reproduced the leading term $h(a_L - a_R)^2/(a_L + a_R)^2$ analytically and confirmed it against $N = 31, \dots, 255$ (`support/verify/check_interface_order.py`, §4.3). Then re-assembled the full two-dimensional layered problem and compared the nodal, midline and maximal face-flux errors with §4.3.	The two-layer statement of §2.2 and the remark on the two-dimensional boundary replaced the single blanket $\mathcal{O}(1)$ claim: first-order nodal and midline flux error, $\mathcal{O}(1)$ maximal face-flux error near the prescribed horizontal boundaries, $\mathcal{O}(1)$ face-conductance error. **Limitation:** the maximum is a boundary effect of the Dirichlet square; it is stated as an observation on this benchmark, not as an estimate valid for other boundary data.
2	Ran the counterexample search and the bound check in `support/verify/check_stability.py`: entrywise ratio for $a = 1 + 9\sin^2(\pi x)$, $\|A_h^{-1}\|_\infty$ against $1/(8a_{\min})$ for five coefficients in one and two dimensions, and the comparison with the constant-coefficient maximum $0.0736 < \frac{1}{8}$ (the remark recording the counterexample, the remark on the constant).	The proof of the convergence theorem was rewritten with a maximum-principle barrier: $A_h\boldsymbol{\varphi} = \mathbf{1}$ exactly for $\varphi = \frac{1}{2}x(1-x)/a_{\min}$ extended by its own boundary values, and $A_h(\boldsymbol{\varphi} - \boldsymbol{w}) = 0$ with $\boldsymbol{\varphi} - \boldsymbol{w} \ge 0$ on $\partial\Omega$. **Limitation:** equality in $\|\boldsymbol{w}\|_\infty \le 1/(8a_{\min})$ holds only in one dimension; in two the constant is an upper bound (0.0736 versus 0.125).

Table 3: Independent verification (continued).

Item	Independent check and evidence location	Final decision / modification / limitation
3	Separated the three diagnostics on the same computed solution: symmetry defect $\|A_h - A_h^T\|/\|A_h\|$, reconstructed two-sided flux jump, and error against the exact flux (§4 and the interface diagnostics of §3; independent re-computation in `test/validate_interface.py`).	The remark on symmetry versus accuracy and §4.3 now report the three numbers side by side instead of treating symmetry as evidence of physical accuracy. **Limitation:** the jump diagnostic is only defined where a two-sided reconstruction exists, so it is not a substitute for an exact solution.

Two further routine checks are recorded for completeness. The smooth benchmark is *blind* to the θ-correction of the face coefficient when $a = 1 + x$: arithmetic and midpoint averaging then produce entrywise identical matrices, so the second-order table in §4.2 cannot validate the correction term (the remark on the θ-correction). And no convergence claim is made anywhere in this report without a named norm and a four-level refinement table; claims about the layered benchmark are exactness statements (the nodal-exactness theorem), not order statements, because $u_* \notin H^2(\Omega)$ there (the remark on the validity of the truncation analysis).

A4. Intellectual ownership declaration

The mathematical formulation, the discretisation, the solvers, the experiments, the validation evidence and the conclusions are the team's own work. Generative AI was used for drafting, clarification and preliminary checking, and every AI-assisted claim was audited as recorded above. Each member affirms that they can explain and modify the team-owned discretisation and solver code, justify the mathematical claims in their own sections, and state the assumptions and the tests applied at each library interface, without AI assistance during the viva. The viva may ask for a small live modification or diagnostic on team-owned work.

Table 4: Ownership declaration. This copy carries the signature of Meichen Ding; the other four rows are completed and signed by each member in their own copy of this table. A declaration is never drafted on another member's behalf.

Name	Signature	Team-owned artefact / contribution	Trusted dependency and interface
Meichen Ding (2362344)	_____	§1.2; §2.1–§2.4; the remarks on the θ-correction, on mesh parity, on the two-dimensional boundary and on symmetry versus accuracy; Appendix A; the scripts in `support/verify/`	`numpy.linalg` dense solves and `scipy.sparse.linalg.eigsh` used only for corroboration: checked against explicit residuals, against the exact layered solution and against a dense Cholesky factorisation

2 Individual Contribution Statement

Part A above is the team log. The statement below is individual: it states in the first person what I did, and it cross-references the audits of Section 1 where the corresponding AI-assisted claim was checked.

Individual Contribution Statement — Project 2 (PDE BVP)

Name: Meichen Ding (2362344)

Role this project: Mathematical theory lead (consistency, adaptivity and convergence): owner of the model formulation, the discrete scheme, its truncation-error and interface analysis, and the discrete-structure results.

The specific things I did this project:

1. Wrote §1.2 and §2.1–§2.4 of the report's `Method.tex`: the continuous problem and its hypotheses, the flux-balance derivation that fixes one shared face coefficient per edge, the index map and block-tridiagonal structure of the assembled operator (with the non-zero count $5n^2 - 4n$), the symmetric Dirichlet elimination and the hand-computed 4×4 matrix at $n = 2$ used as the small-case oracle, the layered exact solution, the second-order local truncation error with its θ-correction, the series-resistance proposition for the interface conductance with the relative error $(1 - c)^2/(4c)$, the SPD / M-matrix / discrete-maximum-principle theorem, and the discrete stability and $\mathcal{O}(h^2)$ convergence theorem with its maximum-principle barrier argument.
2. Wrote the independent verification scripts `support/verify/`: `check_interface_order.py` (one-dimensional conservative scheme, $a_L = 1$, $a_R = 10$, $N = 31, 63, 127, 255$; harmonic nodes exact to 10^{-14} and every face flux equal to $-20/11$, arithmetic nodal error first order with measured orders 1.015/1.008/1.004 and the closed-form chain flux reproduced), `check_stability.py` (the entrywise-ratio counterexample at $a = 1 + 9\sin^2(\pi x)$, ratio 3.10; $\|A_h^{-1}\|_\infty$ against $1/(8a_{\min})$ for five coefficients) and `check_matrix_evidence.py` (symmetry defect, `eigsh` smallest eigenvalue and dense Cholesky on the submitted assembler). These produced the numbers in the appendix tables and in §2.4.
3. Ran the AI-assisted claim audits of Part A: the three-layer separation of the arithmetic-averaging errors (audit 1), the falsification of the Loewner-implies-entrywise argument and its replacement by the barrier proof (audit 2), and the separation of "symmetric and conservative" from "physically correct" (audit 3), including the numerical values that decide each one (0.662 versus 3.85×10^{-13}; $3.10 > 1$; 0.0736 versus 0.125).

What I would do differently: I would pin down the mesh convention with the code owner before writing any convergence sentence. The skeleton uses $h = 1/(n + 1)$ with n interior points while the report uses $h = 1/N$ with N intervals, so the same symbol N means different things in the skeleton, in the layered tables and in the FEM comparison; the interface-parity requirement N odd for the FDM and N even for the FEM then has to be stated as design rather than discovered late. I would also have demanded the two-dimensional layered numbers before writing the one-dimensional flux estimate, because that is exactly where the two-layer statement became necessary.

Anything I should have asked for help with earlier: I should have asked, in Week 5, for each member to commit their own sections and their own ICS from their own account. By the time the appendices were written the repository history was a single upload commit, so authorship had to be reconstructed from memory and from section ownership instead of being visible in `git log`. I also should have asked for a review of the stability proof before it went into a draft: the Loewner-order step survived an internal re-reading and was only caught when it was written out as a counterexample search.

Signature: _____ **Date:** _____